FinTech NLP Engine for E-Satisfaction and Risk Perception in Indian Mobile Banking: A Comparative Twin-Model Sentiment Analysis Using Hierarchical Cascade Architecture

Abstract

Background: Digital banking adoption in emerging economies operates under distinct institutional mandates and user expectations. While technology acceptance literature emphasizes interface usability, operational disruptions in financial applications introduce acute risk perceptions that deteriorate user retention.
Objective: This study models the trade-off between digital satisfaction (E-Satisfaction) and failure-driven anxiety (Risk Perception) across Indian government and private mobile banking platforms.
Methodology: We propose a domain-adapted FinTech Natural Language Processing (NLP) framework trained on a corpus of 1,365,000 user reviews (585,000 government; 780,000 private) extracted from the Google Play Store. To address the semantic noise of code-mixed text (Hinglish) and financial jargon, we deploy a deep learning-ready preprocessing pipeline paired with an IndicBERT encoder. The core architecture uses a Two-Tier Hierarchical Cascade Classifier: Tier 1 filters operational friction from active system failures, and Tier 2 categorizes escalated failures into Technical Performance Risk versus Financial Security Risk.
Expected Results: We evaluate institutional response latency against sentiment recovery metrics, testing whether concrete resolution within 48 hours mitigates user attrition. Comparative analysis measures whether private bank users exhibit lower tolerance for performance latency relative to public sector platform users.

Keywords: FinTech NLP, Sentiment Analysis, Risk Perception, E-Satisfaction, Hierarchical Cascade, IndicBERT, Mobile Banking, Code-Mixed Text.

I. Introduction

The rapid digitalization of financial services in India has shifted primary transaction touchpoints from physical branches to mobile applications. While competitive private sector platforms prioritize UI/UX optimization and hedonic experience, state-operated platforms operate under public service mandates where performance failures directly impair financial access. Existing technology acceptance frameworks—such as the Technology Acceptance Model (TAM) and Information Systems Success Model (ISSM)—predominantly evaluate private sector contexts under normal operating conditions. Consequently, they fail to capture the asymmetric impact of system failures on user risk perception.

Analyzing user feedback in the Indian FinTech domain presents two distinct challenges:

1. **Linguistic Complexity:** Feedback is heavily code-mixed (Hinglish) and populated with domain-specific abbreviations (e.g., UPI, KYC, NEFT, OTP) and localized distress markers (e.g., "paisa kat gaya"). Traditional statistical NLP pipelines (e.g., Bag-of-Words, TF-IDF) fail to capture these contextual semantics.
2. **Label Noise in Unstructured Sentiment:** A single review often aggregates UI complaints, system crashes, and financial anxiety. Standard flat multi-class models fail under these multi-intent conditions, resulting in severe label noise and high false-negative rates for critical security panics.

This paper presents a domain-adapted FinTech NLP engine that addresses these limitations through a Two-Tier Hierarchical Cascade Architecture. By benchmarking 585,000 government bank reviews against 780,000 private bank reviews, we establish a comparative framework for evaluating failure-driven risk dynamics in digital financial platforms.

II. Literature Review & Research Gap

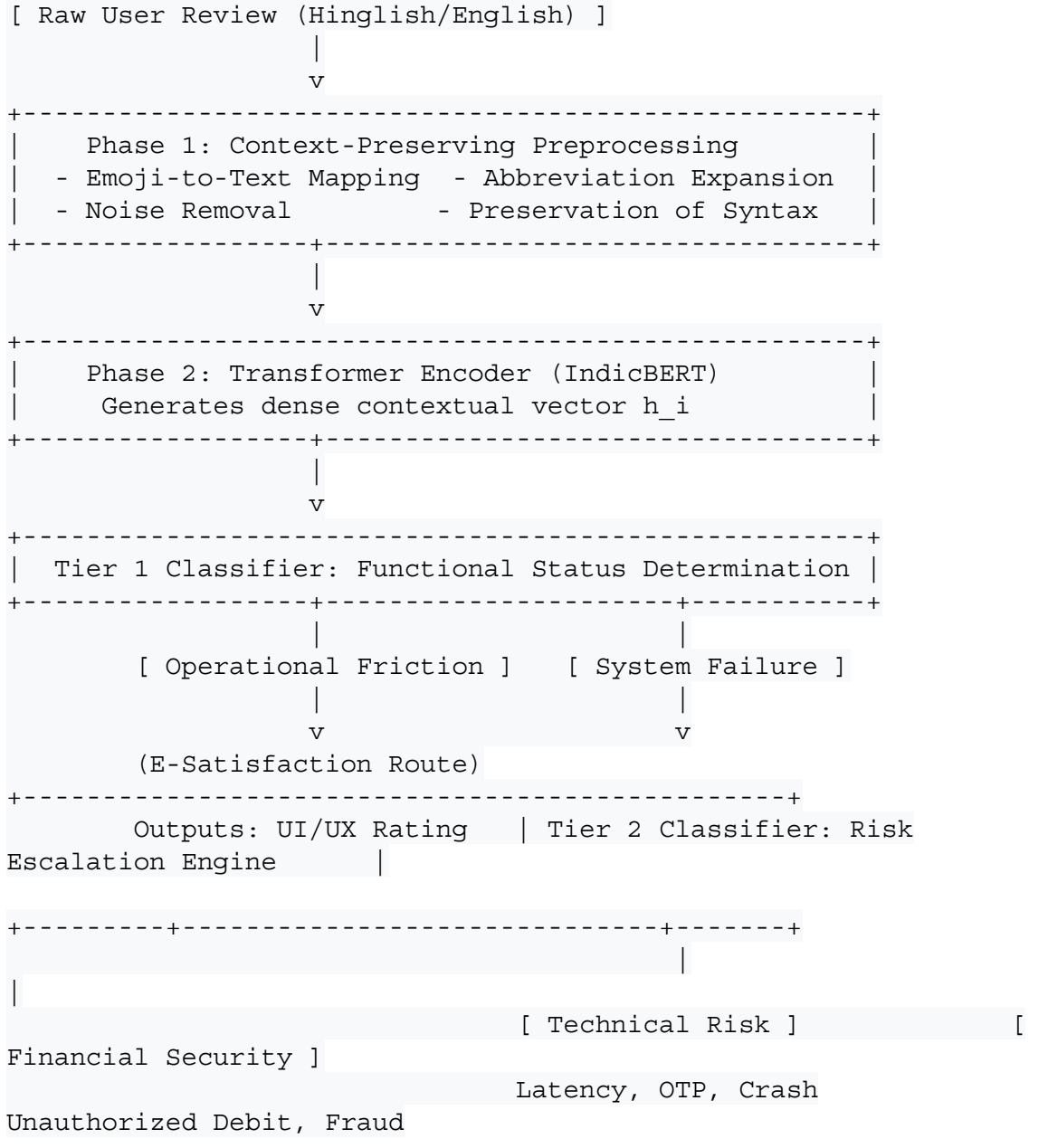
A. Baseline Deconstruction

Recent literature on public sector digital banking (e.g., Cerasi et al., 2026, Technology in Society) demonstrated through 60,000 user reviews of Turkish government banks that **Performance Risk** (e.g., latency, server timeouts) exerts a stronger negative influence on user sentiment than traditional **Security Risk** (e.g., unauthorized access). Furthermore, institutional engagement metrics indicated that concrete responses delivered within a 48-hour window yield significant sentiment recovery.

B. Identified Research Gaps

Dimension	**Baseline Paper (Cerasi et al., 2026)**	**Proposed Framework**
Market Context	Monolingual European/Turkish sector	Code-mixed Indian market (Hinglish + Financial Jargon)
Dataset Scale	N = 60,000 reviews	N = 1,365,000 reviews (585k Public / 780k Private)
Institutional Scope	Public sector banks only	Comparative: Public (e.g., SBI) vs. Private (e.g., HDFC)
Model Architecture	Flat text classification / Topic modeling	Two-Tier Hierarchical Cascade via Transformer Embeddings
Linguistic Processing	Legacy stop-word removal & stemmers	Context-preserving normalization for Transformer models

III. Methodology & System Architecture



A. Corpus Acquisition and Structuring

The dataset comprises 1,365,000 user reviews scraped directly from the Google Play Store across primary Indian banking applications:

- **Government Sector Sub-Corpus:** N = 585,000 reviews (e.g., YONO SBI).
- **Private Sector Sub-Corpus:** N = 780,000 reviews (e.g., HDFC Mobile Banking).

B. Deep Learning-Ready Preprocessing Pipeline

Unlike statistical text models, Transformer architectures (e.g., BERT variants) require complete syntactic structure and positional encodings to compute attention weights. Standard stop-word stripping and aggressive lemmatization degrade Transformer performance. The preprocessing pipeline strictly executes:

1. **Noise Reduction:** Stripping HTML tags, system logs, and non-Unicode artifacts while maintaining sentence boundary markers.
2. **Emoji Tokenization:** Replacing affective emojis with standardized semantic tokens.
3. **Lexicon Normalization:** Expanding domain-specific financial acronyms and code-mixed slang using a specialized lookup mapping.

C. Two-Tier Hierarchical Cascade Classifier

Tier 1: Functional Status Classification

Tier 1 functions as a binary gating mechanism, partitioning input representations into non-critical user friction versus critical system failures. If the probability of a system failure is below the decision threshold, the review is routed to the E-Satisfaction Module. If it is above the threshold, it is flagged as an active system failure and escalated to Tier 2.

Tier 2: Risk Escalation Engine

For reviews escalated to Tier 2, the engine computes categorical risk probabilities over two distinct failure domains: Technical Performance Risk and Financial Security Risk. To mitigate class imbalance (as high-severity financial security panics occur at lower frequencies than routine technical latency complaints), the loss function for Tier 2 incorporates Focal Loss.

IV. Experimental Setup & Evaluation Framework

A. Ground-Truth Annotation Strategy

Given the size of the unlabelled corpus (N = 1.365M), manual annotation of the full dataset is intractable. Ground-truth generation utilizes a **Weak Supervision** pipeline:

1. An open LLM generates initial probabilistic labels across a stratified sample (n = 20,000) using structured zero-shot prompts grounded in the domain taxonomy.
2. A manually annotated validation set (n = 2,000) is used to compute label noise, refine prompt constraints, and calibrate classification boundaries.

B. Evaluation Metrics

Model performance is evaluated using:

- **Macro F1-Score:** To evaluate performance across minority security risk classes.
- **Hamming Loss:** Measuring total misclassified labels across the cascade levels.
- **48-Hour Recovery Delta:** Measuring sentiment shifts in follow-up user ratings relative to

institutional response time.

V. Critical Evaluation & Optimization Vectors

While this architecture improves upon flat classification baselines, two primary technical risks require explicit mitigation during implementation:

1. **Error Propagation in Cascaded Tiers:** A false negative at Tier 1 permanently prevents the sample from reaching the Tier 2 Risk Engine. *Mitigation Vector:* Calibrate the Tier 1 decision threshold to maximize Recall rather than Precision.
2. **Domain Drift in Code-Mixed Text:** Shifting slang conventions in Hinglish can cause performance degradation over time. *Mitigation Vector:* Implement continuous weak supervision loops using domain-adapted transformer embeddings to update classification heads as vocabulary evolves.

VI. Conclusion

This paper presents a scalable, domain-adapted FinTech NLP framework tailored to code-mixed, multi-intent mobile banking reviews. By upgrading flat classification paradigms to a Two-Tier Hierarchical Cascade model and scaling evaluation to 1.36 million reviews across Indian public and private banking sectors, this study bridges theoretical acceptance models with production-grade failure triaging.